J-space Ablation and Externalizing Chain of Thought in Qwen3-4B

1. Hypothesis

I investigate whether Qwen3-4B can compensate for J-space ablation by externalizing its chain of thought (CoT). I hypothesize that external CoT can substitute for storing intermediate states in easy problems because each sub-step (i.e., each forward pass) doesn't require a global workspace; but that more CoT cannot substitute in harder problems where *each forward pass* itself must be difficult. More precisely, I hypothesize that B^*, the external token budget needed to recover 90% of base performance, rises with difficulty, and rises faster than the base model's token usage; and that at some difficulty, recovery fails at any budget.

2. Methodology

I sort problems by difficulty and define B^* as the external token budget required to recover 90% of the base model's performance. My idea is to (1) plot B^* as a function of difficulty, and (2) plot capability recovery with no token budget as a function of difficulty. I do this by estimating logistic curves in difficulty bins then connecting them.

Setup, Ablation, Controls. I use Qwen3-4B (thinking mode). I ablate the model's 'global workspace' with a pre-fitted J-lens: at each token position and each layer, I compute J-lens logits, take the top$-k$ active directions, and subtract the residual stream's projection onto their span.[1] I then calibrate twp ablation strengths: *medium* (layers 22–34, $k = 10$) and *heavy* (layers 22–34, $k = 100$). As a control, I also ablate the span of k random orthonormal directions per layer, with the removal rescaled to match the counterfactual J-space ablation. I also confirm that the ablation does not generically destroy capabilities: at medium dose, closed-book recall moves $20/20 \to\sim 11/20$, while in-context extraction stays $\sim 18/20$. Finally, to test faithfulness to CoT, I truncate completed think traces at 50% and force answers.

Measuring Difficulty. I draw 423 problems from GSM8K (n=150), MATH-500 (n=243 stratified ∼50 per level), and AIME 2024 (n=30). I first run the unablated model 8 times per problem (16,384-token budget) and split these attempts into pile A and pile B. I keep a problem if the clean model solves it in $\geq 2/4$ pile A samples (396/423 kept). I then define *difficulty* for the kept problems as the median think-token count over successful pile-B samples.[2] I then group problems into five difficulty bins (edges 1K/2K/4K/8K think tokens; bin sizes 49/150/110/58/29), subsampled to at most 45 per bin.

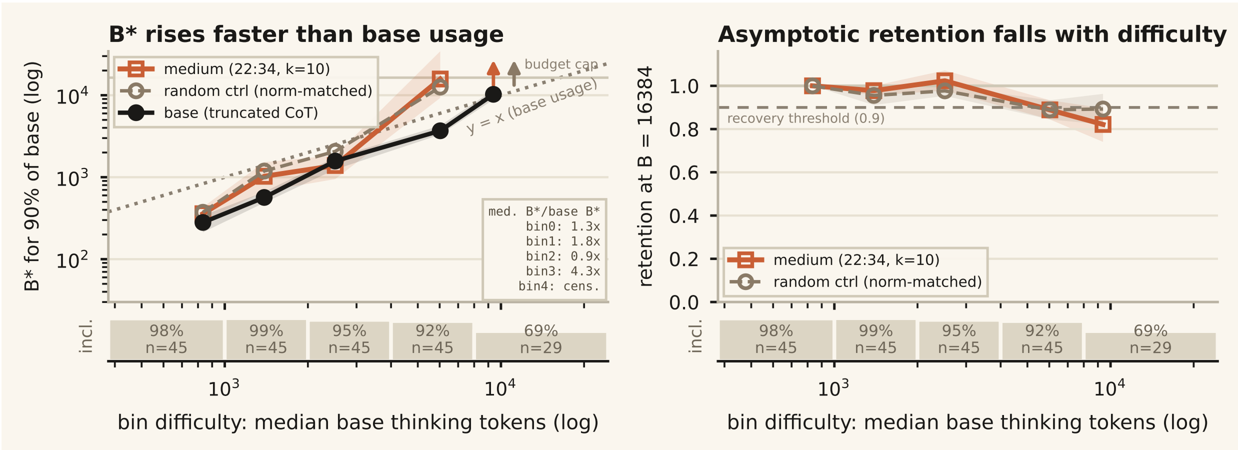
Thinking Budgets. The externalization budget B is the max number of tokens that the model can think for before answering. Concretely, after B think tokens, I force `\n</think>` and supply an answer prefix. I run with $B \in \{0, 128, 512, 2048, 16384\}$ for the base model, medium, and random settings; I run with $B \in \{0, 2048\}$ for the heavy setting due to compute constraints (the model is extremely dumb and reaches the budget nearly always).

Estimation. I count a problem as solved iff $\geq$ 2 of 4 samples solve it. For each difficulty bin and condition, I fit $p(B) = c + (1 - c)\,\sigma(a + b \log(B + 1))$ by MLE, where c is a fitted lower asymptote. I define B^* as the smallest B for which the fitted curve reaches 0.9 times the bin's held-out base rate, and censor B^* if this threshold is not reached by B = 16,384.

[1] Following the paper, I exclude tokens in the top-10 final logits of a clean forward pass, to prevent the ablation from removing the actual next token.

[2] This sample-splitting is to avoid biasing the estimator: if we estimate difficulty on the same sample used for inclusion, we'd bias the estimate downwards since these are runs where the model got 'lucky'.

3. Results



B^* rises with difficulty, and rises faster than the base reference. Figure 1 (left) shows B^* per difficulty bin. If we compare the medium-ablated model to the base model truncated in the same way, the ratio ablated / base across the five difficulty bins is: 1.3x / 1.8x / 0.9x / 4.3x / ∞ (censored).[3] This implies near parity on easy problems, then a sharp divergence above ~4K difficulty.

Recovery fails at high difficulty. In the top bin, the fitted curve never reaches 90% of base by B=16,384. Empirical retention at this budget is 0.82. Bin 3 is borderline ($B^* = 15{,}769$, essentially at the cap). Retention at B=16,384 (Figure 1, right) is: 1.00/0.98/1.02/0.89/0.82 (in increasing order of difficulty).

The control reproduces this behavior. The norm-matched random control is statistically indistinguishable from medium J-space ablation in both B^* and retention. This seems to suggest that ablation generically harms capabilities. However, there are still two interesting points:

1. Although the random vector is norm-matched, the random vector ends up ablating more residual stream content than the medium setting over a trajectory. This is because we compute the J-lens norm on the control's own trajectory, so when we recurrently apply these ablations over a trajectory the two methods can end up ablating different magnitudes (because the J-lens would have e.g. done more on the control trajectory than it actually did). On average, 'medium' removes ~5% of the residual stream's norm, while the control removes 13–15%. Per unit norm, then, it seems like the J-lens is 3x more damaging. With further compute and time, it would be interesting to test whether this rescues the result.
2. The two ablations also behave very differently. At B=16,384, the medium-ablated model hits the cap 59% of the time, while the random control's usage is indistinguishable from the base (3.1% vs. 2.9% of runs hit the cap). It thus seems that J-space ablation destroys the model's ability to make *efficient progress* while random ablation does not.

Heavy ablation dose response. Since heavy ablation only has two budget points, I pool all observations into one logistic.[4] The term that interacts difficulty and token-count (b_3) is null if we fit the same pooled model to medium, which suggests that tokens help at the same rate at every difficulty observed. But for heavy, $b_3 = -0.31 \pm 0.06$ (> 5 SE from zero). This means that the slope is very steep at easy problems (0.77

[3] The base model's B^* is well below the base model's natural usage (e.g., 3,685 vs. a bin-median 6,029 tokens): forcing an early answer often still works, i.e., natural CoT carries roughly 2× slack. Comparing against raw usage would therefore overstate the ablated model's premium.

[4] This is solved like $\sim b_1 \log(B{+}1) + b_2 \log d + b_3 \log(B{+}1) \log d$. The effective budget slope at difficulty d is $b_1 + b_3 \log d$.

in bin 0) and reaches zero at $d \approx 10{,}400$: so the marginal value of an external token declines sharply.

Sanity Checks. I confirm that capabilities are preserved for in-context retrieval and multiple choice questions: MMLU goes 69.0% → 67.6%, and SST-2 goes 86.3% → 84.0%. Next, the difficulty axis is robust: redefining difficulty over all pile-B attempts (including failures) moves $2/396$ problems across bins, the proxy tracks MATH level labels (Spearman $\rho = 0.55$, $p \approx 2 \times 10^{-19}$), and GSM8K vs. MATH problems in the same token-difficulty bin exhibit the same retention. Finally, I truncate CoT traces at 50% and force the model to answer, to test whether tokens are causally significant. Truncating base traces at 50% costs accuracy (0.96 → 0.88), so base CoT is at least somewhat causally load-bearing; but truncating *ablated* traces costs nothing (0.84 → 0.86), suggesting that the model is rambling/has not figured out when to stop.